Active-Rail Spacetime Metric Engineering System
Technical Disclosure and Prototype Plan

Primary Operating Embodiment: Service Factor $V = 5$
High-Stress Operating Embodiment: Service Factor $V = 10$

Document purpose. This document discloses an active-rail spacetime metric engineering architecture, a source-placement control structure, and an engineering prototype plan. The disclosure is organized in a patent-style technical format and emphasizes system components, operating modes, diagnostic procedures, acceptance criteria, and prototype implementation stages.

1. ABSTRACT

An active-rail spacetime metric engineering system is disclosed. The system includes a prepared standing support substrate, a protected live packet corridor, a support-contained radial shift carrier, a locked-lead catch/rematch layer, a radial support-edge softening layer, a lapse-cushion sector, an angular support jacket, and a localized catch-edge absorber. The architecture separates baseline infrastructure support from service-induced transport demand. A dimensionless service factor V specifies the operating load applied to the carried-shift sector of the reduced metric. In a representative operating embodiment at $V = 5$, the system preserves packet causal safety and localizes the service-induced ADM momentum increment primarily into a packet-adjacent catch/rematch handoff channel. A high-stress operating embodiment at $V = 10$ exercises the same architecture near its recognized edge rating. The prototype plan translates the reduced metric fields into ADM variables, computes constraint-ledger fields, subtracts standing substrate demand from live service demand, and screens source-family candidates against localized component roles.

2. TECHNICAL FIELD

The disclosure relates to prescribed spacetime metric engineering, numerical relativity scaffolding, source-placement architectures, and controlled metric-service systems. More particularly, the disclosure relates to a staged rail-like spacetime metric in which prepared infrastructure bears broad geometric support while live service demand is routed through localized handoff windows and packet-adjacent control regions.

3. ENGINEERING OBJECTIVE

A practical spacetime metric engineering architecture benefits from explicit source placement. The disclosed system assigns standing radial support, angular support, lapse safety, carried shift, catch/rematch correction, packet release, and reset operations to distinct service components. The engineering objective is to provide a reduced metric and prototype process that:

1. separates prepared substrate support from live transport increments;
2. keeps the protected packet corridor causally safe throughout live service;
3. localizes service-induced momentum demand into catch/rematch and support handoff regions;
4. expresses the reduced metric in ADM variables for constraint-ledger diagnosis;
5. subtracts the standing substrate from live service fields to isolate transport demand;
6. supports a staged transition from prescribed reduced geometry to a coarse 3+1 constraint prototype.

4. OVERVIEW OF THE ACTIVE-RAIL SYSTEM

The active rail operates as a staged metric service. A prepared standing substrate supplies the baseline throat, support shell, angular jacket, and lapse structure. A live packet corridor lies inside the support shell. A radial shift field carries the service demand through the prepared support rather than through a broad passenger-facing shell. A locked-lead catch/rematch layer absorbs the carried shift and returns the packet toward ordinary local geometry. A compact absorber is applied near the inward packet edge during catch/rematch to address localized handoff demand.

The invention-level structure is the componentized service architecture. The service factor V is an operating-load parameter used to rate embodiments, select support/cushion/absorber settings, and compare source-demand scaling. The core system remains the staged active rail: prepared support substrate, protected packet corridor, support-contained carried shift, locked-lead catch/rematch, packet-adjacent absorber, and resettable infrastructure.

5. DEFINITIONS

Term	Meaning in this disclosure
Active rail	A staged spacetime metric service in which prepared infrastructure supports the transport geometry and a live packet is carried through a protected corridor.
Service factor V	A dimensionless operating-load parameter controlling the magnitude of the support-contained radial shift field during the carried-service portion of the reduced metric. In units with $c = 1$, V is the nominal effective transport factor imposed on the metric service and the load rating used for engineering comparison.
Primary operating embodiment	The representative service configuration evaluated at $V = 5$, selected as the principal prototype target.
High-stress operating embodiment	The representative service configuration evaluated at $V = 10$, selected as an edge-condition test for the same architecture.
Standing substrate	Prepared support geometry including throat/core support, radial support shell, angular jacket, and baseline lapse structure.
Live packet	Passenger-facing corridor whose causal and source-exposure properties are monitored during live service.
Support shell	Wider infrastructure region enclosing the live packet and carrying broad support demand.
Locked-lead catch	Timing rule in which infrastructure-side catch begins slightly before packet rematch while packet rematch remains tightly coupled.
Catch/rematch absorber	Localized packet-edge source or control component assigned to remaining null/momentum handoff demand.
Radial support edge	Transition layer controlling radial pressure demand and support-shell coupling.
Lapse cushion	Lapse-sector compensator that preserves causal margin consumed by stronger support-edge shaping.
Standing substrate subtraction	Diagnostic subtraction of a prepared beta-off or ready-state substrate from live service fields to isolate service-induced ADM demand.

6. OPERATING REGIME AND ROLE OF V

The parameter V is used as a design-rating parameter. Increasing V increases the carried-shift service demand and exposes how the same support geometry responds to higher transport load. The parameter therefore functions as a controlled stress dial for the reduced architecture. It supports comparison across operating embodiments without making the invention depend on a single numerical target.

In representative diagnostics, $V = 5$ provides the primary engineering target. It gives a clean platform for ADM translation, standing-substrate subtraction, catch/rematch source-law development, and source-family screening. The $V = 10$ embodiment exercises the same architecture at a high-stress edge rating. The $V = 10$ case is useful because it intensifies the catch/rematch handoff channel while preserving the same source-placement pattern observed at the primary target.

7. REPRESENTATIVE REDUCED METRIC

A representative reduced metric is expressed as

$$\mathrm{d}s^2 = -\alpha(l,\sigma)^2\mathrm{d}\sigma^2 + A(l,\sigma)\left(\mathrm{d}l + \beta(l,\sigma)\mathrm{d}\sigma\right)^2 + B(l,\sigma)\mathrm{d}\Omega^2, \tag{1}$$

where l is a rail coordinate, σ is a service scheduling parameter, α is the lapse, $\beta = \beta^l$ is the radial shift, $A = \gamma_{ll}$ is the radial spatial metric component, and $B = \gamma_{\Omega\Omega}$ is the angular spatial metric component.

Representative parameter values for one embodiment are listed in Table 2. The values define a prototype design point and may be varied according to engineering load rating, source-family selection, grid resolution, or constraint-solve requirements.

Parameter	Representative value
Primary service factor	$V = 5$
High-stress service factor	$V = 10$
Packet radius	$R_{\text{pass}} = 0.35$
Support radius	$R_{\text{th}} = 1.75$
Angular jacket radius	$R_{\Omega} = 1.75$
Radial support width	$w_{\text{th}} = 0.569$
Packet transition width	$w_{\text{pass}} = 0.06$
Angular jacket width	$w_{\Omega} = 1.4$
Angular jacket amplitude	$a_{\Omega} = 0.20$
Lapse cushion strength	$\eta_N = 2.00$
Support catch center and width	$x = -0.05$, $w = 0.32$
Packet rematch center and width	$x = 0.00$, $w = 0.32$
Catch/rematch profile	minimum-jerk

Table 2: Representative active-rail parameters.

8. SERVICE SEQUENCE

A representative service cycle proceeds through the following phases:

$$\begin{gathered}\text{prepared substrate} \to \text{ready support plant} \to \text{packet entry} \to \text{locked-lead catch/rematch} \\ \to \text{support-contained shift handoff} \to \text{packet release} \to \text{infrastructure reset.}\end{gathered} \tag{2}$$

Passenger-facing accounting is assigned to packet entry, catch/rematch, carried service, release, and any selected live buffer window. Reset and decompression are infrastructure operations unless a diagnostic window intentionally includes them for system-level energy accounting.

9. ADM TRANSLATION

The metric of Eq. (1) is expressed in ADM form through

$$\gamma_{ij} = \mathrm{diag}\left(A, B, B\sin^2\theta\right), \qquad \beta^i = (\beta, 0, 0), \qquad \beta_i = (A\beta, 0, 0). \tag{3}$$

The unit normal to a $\sigma =$ constant slice is

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta}{\alpha}, 0, 0\right), \qquad n_\mu = (-\alpha, 0, 0, 0). \tag{4}$$

Using dot notation for ∂_σ and prime notation for ∂_l, and adopting the convention

$$K_{ij} = \frac{1}{2\alpha}\left(D_i\beta_j + D_j\beta_i - \partial_\sigma\gamma_{ij}\right), \tag{5}$$

the independent extrinsic-curvature components are

$$K_{ll} = \frac{1}{2\alpha}\left(2A\beta' + A'\beta - \dot{A}\right), \tag{6}$$

$$K_{\theta\theta} = \frac{1}{2\alpha}\left(\beta B' - \dot{B}\right), \tag{7}$$

$$K_{\phi\phi} = K_{\theta\theta}\sin^2\theta. \tag{8}$$

The mixed eigenvalue-like quantities are

$$k_l = K^l{}_l = \frac{1}{2\alpha}\left(2\beta' + \beta\frac{A'}{A} - \frac{\dot{A}}{A}\right), \tag{9}$$

$$k_\Omega = K^\theta{}_\theta = K^\phi{}_\phi = \frac{1}{2\alpha}\left(\beta\frac{B'}{B} - \frac{\dot{B}}{B}\right). \tag{10}$$

Thus

$$K = k_l + 2k_\Omega, \qquad K_{ij}K^{ij} = k_l^2 + 2k_\Omega^2. \tag{11}$$

For spatial line element $\mathrm{d}l_3^2 = A\mathrm{d}l^2 + B\mathrm{d}\Omega^2$, the scalar curvature is

$${}^{(3)}R = -\frac{2B''}{AB} + \frac{(B')^2}{2AB^2} + \frac{A'B'}{A^2B} + \frac{2}{B}. \tag{12}$$

The Hamiltonian and radial momentum constraints produce the ADM ledger fields

$$16\pi\rho_{\mathrm{ADM}} = {}^{(3)}R + K^2 - K_{ij}K^{ij} = {}^{(3)}R + 4k_lk_\Omega + 2k_\Omega^2, \tag{13}$$

$$8\pi j_l = -2k_\Omega' + \frac{B'}{B}(k_l - k_\Omega), \tag{14}$$

$$j_\theta = 0, \qquad j_\phi = 0. \tag{15}$$

10. SOURCE-PLACEMENT LEDGER

The source-placement ledger assigns demanded source roles to system components. The ledger is used for engineering localization, source-family screening, and prototype acceptance.

Source role	Engineering assignment	Primary diagnostic
Standing support shell	Prepared substrate and throat/core support	baseline ρ_{ADM}
Radial support edge	Radial pressure reduction and support-shell shaping	edge gradients
Angular jacket	Broad angular/effective support	angular support burden
Lapse cushion	Causal-margin compensator	packet norm margin
Shift packet carrier	Support-contained carrying field	momentum localization
Catch/rematch absorber	Inner packet-edge handoff absorber	Δj_l, T_{kk} residual
Release/fade correction	Controlled shift fade after packet normalization	release-edge leakage
Reset/decompression infrastructure	Reset after live service	infrastructure ledger

The representative diagnostics indicate that the live service increment appears primarily in j_l during catch/rematch. The absolute ρ_{ADM} activity within the packet corridor is substantially attributable to standing substrate curvature, as demonstrated by standing substrate subtraction.

11. CATCH-EDGE CONTROL LAW

Let

$$\xi = l - s(\sigma) \tag{16}$$

be a packet-centered radial coordinate. A preferred absorber amplitude is a compact kinematic handoff law multiplied by an inner-edge envelope:

$$A_{\text{catch}}(\xi, \sigma) = W_{\text{catch}}(\sigma)\left[a\, H_{\text{shift}}(\xi, \sigma) + b\, B_{\text{inner}}(\xi)\right], \tag{17}$$

where W_{catch} is a locked-lead timing window, H_{shift} is a smooth shift-handoff or shift-mismatch measure, and B_{inner} is a compact bump centered near $\xi = -R_{\text{pass}}$. A residual-responsive limiter may bound the amplitude:

$$A_{\text{catch}} \leq C\left([R_{T_{kk}}]_+\right), \tag{18}$$

where C denotes a smooth cap and $R_{T_{kk}}$ denotes the predicted radial-null residual. The primary control law is kinematic; the residual-responsive term supplies protective limiting and diagnostic alignment.

12. STANDING SUBSTRATE SUBTRACTION

Standing substrate subtraction isolates service-induced ADM demand from prepared infrastructure demand. For a live service field F_{live} and a time-matched prepared beta-off substrate field F_{sub}, define

$$\Delta\rho_{\text{ADM}} = \rho_{\text{ADM,live}} - \rho_{\text{ADM,sub}}, \tag{19}$$

$$\Delta j_l = j_{l,\text{live}} - j_{l,\text{sub}}. \tag{20}$$

This diagnostic determines whether packet-region ADM activity belongs to baseline support curvature or to the active service increment.

In the representative $V = 5$ primary embodiment, exact-builder ADM diagnostics give the results in Table 4.

Observable	Representative diagnostic result
Positive live packet norm samples	0
Maximum live packet norm	-180.546
Live packet fraction of $\lvert j_l \rvert$	7.85%
Live packet fraction of $\lvert \rho_{\mathrm{ADM}} \rvert$	27.06%
Live packet fraction of negative ρ_{ADM}	0.00%
Subtracted packet $\Delta\rho$ relative to live packet ρ	about 0.000295%
Subtracted live-packet Δj_l fraction	about 11.75%

Table 4: Representative $V = 5$ ADM and substrate-subtraction diagnostics.

In the $V = 10$ high-stress embodiment, the subtracted $\Delta\rho$ remains small while catch/rematch-localized Δj_l increases to about 19.25% of live-packet j_l. Approximately 97.94% of the live-packet Δj_l burden is localized in catch/rematch. The comparison supports use of $V = 5$ as the primary prototype target and $V = 10$ as the edge-condition test.

13. PROTOTYPE PLAN

13.1 Prototype Objective

The prototype objective is to construct a coarse ADM diagnostic scaffold for the $V = 5$ active-rail service and determine whether the prescribed reduced metric produces a constraint-readable source-placement target. The prototype begins with prescribed geometry, then proceeds toward constraint-compatible source modeling.

13.2 Stage 1: Exact Field Builder and ADM Ledger

Inputs:

- exact field functions $\alpha(l, \sigma)$, $\beta(l, \sigma)$, $A(l, \sigma)$, and $B(l, \sigma)$;
- packet, support, stage, and region labels;
- primary $V = 5$ operating target and high-stress $V = 10$ comparison case.

Outputs:

- maps of k_l, k_Ω, K, ${}^{(3)}R$, ρ_{ADM}, and j_l;
- region and stage burden tables;
- packet norm and packet exposure diagnostics.

Acceptance criteria:

1. all live-packet norm samples remain negative;
2. ADM fields remain finite on the diagnostic grid;
3. live momentum burden concentrates in support/catch/rematch rather than packet interior;
4. live negative ρ_{ADM} packet burden remains zero or negligible.

13.3 Stage 2: Standing Substrate Subtraction

Compute beta-off or ready-state substrate fields and subtract them from live service fields. The objective is to identify the service-induced $\Delta\rho$ and Δj_l increments.

Acceptance criteria:

1. $\Delta\rho$ remains small inside the live packet;
2. Δj_l is localized near catch/rematch and support-side handoff regions;
3. the $V = 10$ case increases handoff demand while preserving the qualitative localization pattern observed at $V = 5$.

13.4 Stage 3: Compact Source-Control Law

Replace fitted modal catch-edge components with a compact control law. The preferred embodiment uses a kinematic handoff amplitude and an inner-edge spatial envelope. The law is tested against residual maps and ADM Δj_l maps.

Acceptance criteria:

1. the law captures the majority of catch/rematch Δj_l demand at $V = 5$;
2. the support remains localized near $\xi = -R_{\text{pass}}$;
3. packet-region $\Delta\rho$ remains small;
4. scaling from $V = 5$ toward $V = 10$ remains smooth.

13.5 Stage 4: Coarse 3+1 Constraint Prototype

A coarse 3+1 prototype imposes or solves partial constraint data using the ADM scaffold as the target. Candidate subtests include:

- fixing γ_{ij} and computing compatible source ledgers;
- smoothing or regularizing the catch-edge Δj_l layer;
- comparing beta-off, low-beta, and full-beta substrate choices;
- testing convergence of $\Delta\rho$ and Δj_l under grid refinement;
- repeating the acceptance suite at $V = 10$ as an edge-condition comparison.

13.6 Stage 5: Source-Family Screening

Candidate source families are screened against the component ledger. Source families may be assigned to standing support, angular jacket, lapse support, shift carrier, and catch-edge absorber roles. Preference is given to source families that keep amplitudes bounded, preserve packet-region safety, and maintain localized support roles under refinement.

14. ENGINEERING ACCEPTANCE CRITERIA

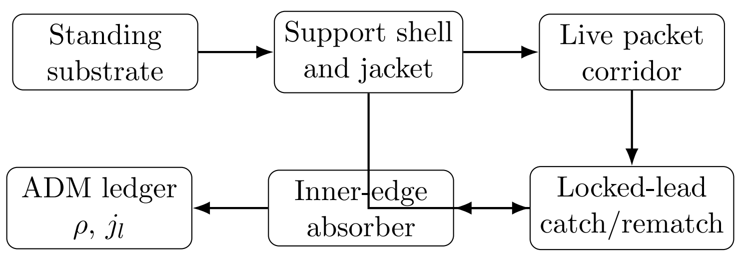
Criterion	**Meaning**	**Gate**
Packet norm safety	All live packet norm samples remain negative	Required
Substrate-separated density	Packet $\Delta\rho$ remains small after subtraction	Required
Handoff localization	Service Δj_l concentrates in catch/rematch and support handoff	Required
Residual localization	Residual maps show localized packet-edge and support-region structure	Required

$V = 5$ stability	Primary target remains stable under grid and seed variation	Required
$V = 10$ edge behavior	High-stress case intensifies the handoff channel while preserving source placement	Desired
Compact absorber law	Catch-edge absorber is expressible as a bounded control law	Required
Constraint compatibility	Coarse 3+1 prototype maintains stable constraint behavior	Required

15. RISK CONTROLS AND DIAGNOSTIC ALARMS

1. **Packet-region density increment.** A large $\Delta\rho$ concentration inside the protected packet would trigger packet-corridor redesign.
2. **Refinement-sharpening catch layer.** A catch-edge layer that sharpens under grid refinement would trigger absorber broadening or source-family rejection.
3. **Angular jacket leakage.** Angular support entering the packet region would trigger redesign of the angular jacket or support-shell partition.
4. **Momentum delocalization.** Δj_l spreading across the packet interior would trigger revision of the handoff model.
5. **High-stress qualitative transition.** A $V = 10$ source pattern that changes category rather than magnitude would place the edge case outside the primary prototype rating.
6. **Constraint instability.** Growing constraint residuals in the coarse 3+1 scaffold would trigger smoothing, reparameterization, or source-ledger redesign.

16. ILLUSTRATIVE SYSTEM DIAGRAM



17. CLAIM-ORIENTED EMBODIMENTS

The following claim-oriented embodiments clarify the technical scope of the disclosure.

1. A method of operating a prescribed spacetime metric service, comprising preparing a standing support substrate, defining a live packet corridor within a larger support shell, routing a radial shift field through the support shell, and applying a locked-lead catch/rematch control window to localize service-induced momentum demand near a packet edge.
2. The method of item 1, wherein the live packet corridor has radius R_{pass} smaller than a support radius R_{th}, and wherein a radial support-edge width w_{th} is selected to reduce radial pressure burden while a lapse cushion preserves packet causal margin.
3. The method of item 1, wherein a dimensionless service factor V controls the carried-shift operating load, and wherein V is used to select support, lapse-cushion, and catch/rematch absorber settings.
4. The method of item 3, wherein a primary operating embodiment is rated at $V = 5$ and a high-stress operating embodiment is rated at $V = 10$.

5. The method of item 1, wherein an ADM diagnostic ledger is computed from fields α, β, $A = \gamma_{ll}$, and $B = \gamma_{\Omega\Omega}$, and wherein the ledger includes ρ_{ADM} and j_l.
6. The method of item 5, further comprising subtracting a time-matched standing substrate from a live service field to identify $\Delta\rho$ and Δj_l.
7. The method of item 6, wherein a service pass is accepted when $\Delta\rho$ remains small inside the live packet and Δj_l is localized near a catch/rematch handoff layer.
8. A spacetime metric engineering system comprising a standing support shell, an angular jacket, a lapse-cushion sector, a support-contained shift carrier, a live packet corridor, and a packet-edge catch/rematch absorber.
9. The system of item 8, wherein the packet-edge catch/rematch absorber comprises a kinematic handoff law multiplied by a compact inner-edge spatial envelope centered near $\xi = -R_{\mathrm{pass}}$.
10. The system of item 9, wherein a residual-responsive limiter bounds the absorber amplitude.
11. A numerical prototype process comprising constructing an exact field builder for α, β, A, and B; computing ADM constraint fields; performing standing substrate subtraction; and screening source-family candidates against component-specific source roles.

18. PROTOTYPE DELIVERABLES

The prototype deliverables for the primary $V = 5$ operating embodiment are:

1. exact-builder ADM implementation package;
2. substrate-subtracted ρ_{ADM} and j_l maps;
3. catch-edge absorber law specification;
4. $V = 5$ acceptance report;
5. $V = 10$ high-stress comparison appendix;
6. source-family screening matrix;
7. coarse 3+1 constraint prototype specification.

19. CONCLUSION

The active-rail system organizes spacetime metric engineering as a source-placement architecture. The primary $V = 5$ operating embodiment separates standing substrate curvature from live service demand and localizes the principal service increment into a packet-adjacent catch/rematch momentum channel. The $V = 10$ operating embodiment supplies an edge-condition comparison for the same architecture. The engineering path proceeds through exact-builder ADM diagnostics, standing substrate subtraction, compact absorber-law specification, source-family screening, and coarse 3+1 constraint prototyping.